# Eric Peters

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## Professional Profile

MIT-educated structural analyst with 15 years of specialized experience in the design and analysis of satellites, launch vehicles, and human-rated spacecraft. Expertise spans composite design, advanced simulation (multi-body dynamics; nonlinear structural FEA), and damage tolerance analysis. Currently expanding into software development, leveraging modern web technologies to build engineering analysis tools with enhanced user experience. Eager to apply this unique combination of technical expertise and innovative spirit to challenges in aerospace and renewable energy industries.

## Education

**Massachusetts Institute of Technology** — Cambridge, MA

Master of Science, Aerospace Engineering — 2012—2014

- Thesis: *"Dynamic Instabilities Imparted by CubeSat Deployable Solar Panels"*

Bachelor of Science, Aerospace Engineering — 2007—2011

## Experience

**Blue Origin** — Kent, WA

*Senior Structural Analyst* — 2024—Present

- Owned end-to-end analysis of the New Shepard Crew Capsule (CC) Escape subsystem, conducting nonlinear structural and random vibration analyses to validate solid rocket motor (SRM) block upgrades.
- Led critical damage tolerance initiatives for Block 1 and Block 2 SRM systems, delivering six analysis packages that eliminated a long-standing waiver and proved hardware survivability for the vehicle's required life.
- Directed the redesign and validation of the CC separation spring following a flight anomaly, performing a rapid impact dynamics analysis that enabled expedited testing and integration of new hardware before the next mission.
- Authored means of compliance document for CC Structures subsystem block upgrade; defined analytical methods, verification approaches, and component-specific safety factors to satisfy over 100 structural requirements for human flight certification.
- Developed Python automation scripts to streamline safety factor selection, load case combination, and post-processing of fastener forces from FEA models, significantly reducing manual analysis time and improving data consistency.
- Collaborated closely with subcontractor analysts, reviewing submissions and mandating improvements to designs and methods to align all third-party deliverables with stringent company standards.

*Structures Design Engineer* — 2016—2023

- Authored over 20 comprehensive analysis packages for safety-critical metallic and composite components within CC Structures subsystem, supporting the initial human flight certification process that culminated in the inaugural crewed flight in July 2021.
- Redesigned the aft primary beam assembly for CC2.1 increment, leveraging lessons learned from certification process to simplify load paths and increase strength margins by 40% while reducing subsystem mass by 10%.
- Sourced and maintained supplier relationships for high-value composite components (>$100k unit cost), ensuring compliance with AS9100 quality standards.
- Defined and implemented repair procedures for manufacturing and flight operations groups, resolving 300+ unique nonconformances across three vehicles and achieving program goal of a 50% reduction in turnaround time between flights.

**Freelance - Spacecraft Systems Architect** — 2021

- Conducted a three-month conceptual design study for an ESPA-class weather radar satellite, culminating in the delivery of subsystem sizing tools, preliminary technical budgets, and a report comparing the merits of three architectural layouts against top-level mission requirements for ground coverage and mass/volume constraints.

**Firefly Space Systems** — Cedar Park, TX

*Lead Design Engineer, Payload Segment Structures* — 2014—2016

- Led a team of three engineers to design and analyze payload fairing, payload attachment structures, and associated manufacturing tooling for the Alpha 1.0 launch vehicle.
- Instituted an elementary systems engineering process tailored around limited personnel and software resources to aid development of Design Reference Missions, technical budgets, and subsystem functional requirements.
- Cultivated the initial relationship between Firefly executive team and Seedinvest, an equity crowdfunding platform, that resulted in over $1 million of seed round funding.
- Authored and maintained payload accommodations sections of the Firefly Alpha Payload User's Guide. Coordinated with customers to define mechanical and electrical interfaces, payload integration facility requirements, and multi-payload deployment CONOPS.

**MIT Space Systems Laboratory** Cambridge, MA
*Undergraduate / Staff / Graduate Researcher* 2010—2014

- Designed motor assembly and chassis for Micro-sized Microwave Atmospheric Satellite (MicroMAS) 3U weather-monitoring CubeSat. Supported hardware fabrication, vehicle integration, and qualification/acceptance testing of flight hardware. Demonstration missions MicroMAS-1 and MicroMAS-2 launched in May 2014 and January 2018.
- Matured design of Regolith X-ray Imaging Spectrometer (REXIS) primary structure from initial concept to PDR fidelity. Sized detector electronics shielding to protect against space radiation environment using results from ESA SPENVIS model. Launched aboard NASA OSIRIS-REx mission in September 2016.

## Skills

**Computer-Aided Design (CAD):** Autodesk Inventor, Fusion; Dassault CATIA, SolidWorks; PTC Creo
**Finite-Element Analysis (FEA):** Altair HyperMesh, Optistruct, RADIOSS; Ansys Mechanical; MSC Adams; NASGRO; Siemens Femap/NASTRAN
**Technical Skills:** Matlab; Python; JavaScript/React/Vue; Docker; Linux

## Projects

**EPeters.io** | Website

- Personal website showcasing portfolio of engineering tools along with blog posts inspired by niche technical topics encountered throughout my career.

**Enginuity** | Website

- Unified platform for structural analysis tools.

**Galvanic Compatibility Visualizer** | Website

- Visualize the potential for galvanic corrosion between pairs of dissimilar metals and recommended surface treatments based on MIL-STD-889-C data.

**Stressed** | Website

- Plot Mohr's Circle for a given 2D or 3D stress state and calculate principal, Tresca, and von Mises stresses.

**Self-Hosted LLM Knowledgebase** | *Unpublished*

- Implemented a self-hosted, automated platform using multi-modal LLMs for document summarization and keyword extraction to support personal knowledge management database, showcasing skills in development and integration of custom AI workflow solutions.

## Publications


1. Blackwell, William et al. (2013). “MicroMAS: A first step towards a nanosatellite constellation for global storm observation”. In: Proceedings of the AIAA/USU Conference on Small Satellites. Around the Corner, SSC13-XI-1.
2. Peters, Eric (2012). “Challenges of Mechanism Design for Small Educational Satellites”. In: Mechanical Engineering Technology Symposium. Lexington, MA: MIT Lincoln Laboratory.
3. Peters, Eric et al. (2014). “Design and functional validation of a mechanism for dual-spinning cubesats”. In: The 42nd Aerospace Mechanism Symposium. NASA Goddard Space Flight Center.